C.S. MOTT CHILDREN'S HOSPITAL
MICHIGAN MEDICINE

MANAGEMENT OF THE NEONATE WITH PERINATAL HIV EXPOSURE
UNIVERSITY OF MICHIGAN CLINICAL PRACTICE GUIDELINE

I. OVERVIEW:

Human immunodeficiency virus (HIV) is a retrovirus that infects humans and may result in significant effects on immune function, potentially leading to the development of acquired immune deficiency syndrome (AIDS). The management of neonates with known or possible perinatal HIV exposure requires the immediate initiation of antiretroviral drugs for prophylaxis. The use of antiretroviral drugs reduces perinatal transmission by several mechanisms, including lowering parental antepartum viral load and providing infant pre- and post-exposure prophylaxis. Therefore, combined antepartum, intrapartum, and neonatal antiretroviral prophylaxis is recommended to prevent perinatal transmission of HIV.

These guidelines are adapted from recommendations outlined by the Panel on Treatment of HIV During Pregnancy and Prevention of Perinatal Transmission, updated January 2023.

II. PURPOSE:

The purpose of this document is to guide clinicians through the immediate postnatal management of neonates with known or possible perinatal HIV exposures.

III. SCOPE:

This scope of this document encompasses the initial management (including diagnostic testing, antiretroviral prophylaxis, and necessary consultation and follow-up) for an infant born to an individual with either a known positive HIV status or unknown HIV status and cared for in the newborn nursery or the NICU.

IV. DEFINITIONS:

Perinatal HIV Exposure: Exposure occurs when an infant is born, via any mode of delivery, to an individual who is positive for HIV. The infant is considered to have perinatal HIV exposure regardless of the birthing parent's burden of disease (i.e., their HIV viral load), whether the birthing parent received antepartum/intrapartum antiretroviral therapy, or the mode of delivery.

Antepartum antiretroviral drugs: Antiretroviral drugs given to the birthing parent during pregnancy.

Intrapartum antiretroviral drugs: Antiretroviral drugs given to the birthing parent at the time of delivery.

Antiretroviral Prophylaxis: The administration of one or more antiretroviral drugs to a newborn without documented HIV infection to reduce the risk of perinatal acquisition of HIV.

Presumptive HIV Therapy: The administration of a three-drug antiretroviral regimen to newborns who are at highest risk of perinatal acquisition of HIV. Presumptive HIV therapy is intended to be preliminary treatment for a newborn who is later documented to have HIV, but it also serves as prophylaxis against HIV acquisition for those newborns who are exposed to HIV in utero, during the birthing process, or during breastfeeding and who do not acquire HIV.

Low-risk infants: Infants born to individuals with known HIV who are currently receiving and have received at least ten consecutive weeks of antiretroviral therapy during pregnancy, have achieved and maintained viral suppression (defined as two consecutive tests with HIV RNA level <50 copies/ml obtained at least four weeks apart), have a viral load <50 copies/ml at or after 36 weeks gestation, did not have acute HIV infection during pregnancy, and report good adherence to antiretroviral therapy, with no adherence concerns identified.

High-risk infants: Infants born to individuals with known HIV who have not received any antepartum or intrapartum antiretroviral drugs, have received only intrapartum antiretroviral drugs, have received appropriate antepartum/intrapartum antiretroviral drugs but did not achieve viral suppression (defined as two consecutive tests

with HIV RNA level <50 copies/ml obtained at least four weeks apart) within four weeks of delivery, or who had primary/acute HIV infection during pregnancy.

V. GUIDELINE:

Prenatal Management

- We highly recommend that all pregnant individuals with HIV be referred to Pediatric Infectious Diseases for a prenatal consult visit to discuss plans for infant treatment.
    - Visit will be documented in the chart and in a care coordination note
    - Patient will be provided with printed documentation of the visit in case they deliver outside of Michigan Medicine
- Patient-centered counseling to support shared decision making about infant feeding (e.g. breast feeding versus formula feeding) should begin as early as possible in pregnancy.

Immediate Postnatal Management of the HIV-Exposed Neonate

Consults

- Please consult Pediatric Infectious Diseases when an infant is born to an individual with known HIV infection, or when an infant is born to an individual with unknown HIV status who subsequently tests positive for HIV. This will allow the Pediatric Infectious Diseases service to establish care and answer questions for the family. Do not delay starting antiretroviral prophylaxis while waiting for the Pediatric Infectious Diseases consult to be completed.
- Pediatric Infectious Diseases must still be consulted after birth even if a prenatal visit has been done.
- Social work should be consulted to evaluate infants with HIV infection and their families prior to discharge.

Antiretroviral Prophylaxis

- Antiretroviral prophylaxis should be initiated **as soon as possible after birth** for all HIV-exposed neonates. Ideally, the first dose should be given within 6 hours of delivery and no later than 12 hours after delivery (see recommendations for regimens and dosing in Tables 1 and 2).

- **Infants at low risk of HIV transmission** should receive antiretroviral prophylaxis with a course of zidovudine.
    - Dosing recommendations by gestational age at birth (see Table 2)
    - Duration of therapy by gestational age at birth (see Table 2)
        - Born less than 37 weeks: Receive 4-6 week course
        - Born at or after 37 weeks: Receive 2 week course

- Infants who do not meet all criteria for being low risk for transmission but whose birthing parent had a viral load of 50 copies/ml or less after 36 weeks gestation should receive zidovudine for 4-6 weeks.

- **Infants at high risk of HIV transmission** should receive **presumptive HIV therapy with a three-drug regimen**: zidovudine, lamivudine, and either nevirapine or raltegravir (see recommended regimens in Table 1 and dosing recommendations in Table 2).
    - For high-risk infants receiving presumptive therapy, zidovudine should be continued for six weeks. The duration of lamivudine and nevirapine or raltegravir depends on a variety of factors and will be determined during outpatient follow-up in the Pediatric Infectious Diseases Clinic.

- There is no consensus on appropriate management of antiretroviral prophylaxis for infants of individuals with sustained viral suppression who are breastfed. Most experts recommend between 2-6 weeks of zidovudine. The Pediatric Infectious Diseases service will guide antiretroviral management for these patients. In general, we recommend prophylaxis with zidovudine for a minimum of two weeks and until breastfeeding has been established using the practices to mitigate risk listed below.

- The recommendations for neonatal prophylaxis regimens described above still apply to infants born to individuals with HIV infection and known antiretroviral resistance.

- In cases where there is high risk of transmission, the physician managing the birthing parent's HIV treatment should contact Pediatric Infectious Diseases, preferably prior to delivery, to discuss optimal antiretroviral prophylaxis for the newborn in greater detail.
    - The National Perinatal HIV Hotline (1-888-448-8765) provides free 24-hour clinical consultation on all aspects of perinatal HIV, including infant care, and may also be a helpful resource in this situation.

Infant Feeding

- Individuals with HIV should receive evidence-based, patient-centered counseling to support shared decision making about infant feeding.

- Individuals with HIV who are on ART with a sustained undetectable viral load and who choose to breastfeed should be supported in this decision.

- Individuals who decide to breastfeed their infants should be counseled on strategies to mitigate the risk of transmission of HIV via breastmilk, including the following:
    - Referral to a lactation consultant is recommended, including outpatient follow up within the first week after discharge
    - Exclusive breastfeeding up to six months of age is recommended over mixed feeding (i.e. breastmilk and formula), acknowledging that there may be an intermittent need to give formula.
    - Solids should be introduced at six months of age and not earlier.
    - The breastfeeding parent should have their viral load regularly monitored (e.g. every 1-2 months). If viral load becomes detectable, breastfeeding should be stopped.
    - Mastitis, thrush, and cracked or bleeding nipples should be promptly identified and treated. Milk from the affected breast should be pumped and either discarded or flash heated prior to giving to infant; milk can be fed as usual from the unaffected breast.
    - When ready to wean the infant from breastmilk, this should be done carefully and over 2-4 weeks to prevent engorgement and milk stasis.

- Replacement feeding with formula or donor breast milk is recommended for infants of individuals with HIV who are not on ART and/or do not have a suppressed viral load during pregnancy (at a minimum throughout the third trimester) and at delivery. All patients with HIV have access to donor milk.

Laboratory Evaluation:

- For high-risk infants, complete blood count and differential (CBCPD) and HIV RNA PCR should be obtained at birth.

- For low-risk infants who are being breastfed, CBCPD and HIV RNA PCR should be obtained at birth.

- Low-risk infants who are not being breastfed do not require any laboratory evaluation at birth.

Discharge Preparation

- Prescriptions for the infant's antiretroviral medications must be sent to the Taubman Outpatient Pharmacy and the medications must be in parents'/guardians' hands prior to infant's discharge from the hospital. Prescriptions should not be sent to other outpatient pharmacies (e.g. CVS, Walgreens), because liquid formulations of these medications are frequently not in stock.

- Medication administration should be reviewed with parents/guardians in detail prior to discharge to ensure that they are able to reliably give medication to the infant. Parents/guardians are highly encouraged to administer at least one dose under the supervision of the nurse prior to discharge.

- Please contact the Pediatric Infectious Diseases consult team prior to discharge to confirm that a follow-up appointment has been scheduled.

Special Considerations

Infants Born to Individuals with Unknown HIV Infection Status

- For birthing parents with unknown HIV status, rapid HIV-1/2 antigen/antibody testing of the birthing parent and/or infant is recommended as soon as possible.

- Rapid testing should be performed on the birthing parent on admission/prior to delivery. If testing was not performed prior to delivery, it should be performed as soon as possible on the birthing parent or infant. This order is found in MiChart under "Rapid HIV" and has a turn-around time of 60 minutes. For the infant, only 3-4 drops of blood are required and can likely be obtained via heel stick. The test is not validated on cord blood.

- In the setting of a positive parental rapid HIV test, confirmatory testing with HIV RNA PCR should be performed on the birthing parent as soon as possible in addition to HIV-1/2 antibody differentiation assay which is performed as a reflex test of rapid test, and the infant should immediately be started on three-drug presumptive HIV therapy. CBC with differential and HIV RNA PCR should be performed on the infant.
    - If the parental HIV RNA PCR is negative, then the infant's antiretroviral prophylaxis should be discontinued.
    - If the parental HIV RNA PCR is positive, three-drug presumptive HIV therapy should be continued for the infant and the management should follow that of high-risk exposure.
    - If the birthing parent is unavailable or declines confirmatory testing, a positive HIV test result should be assumed. The infant should receive three-drug presumptive HIV therapy and the management should follow that of high risk exposure.

- In the setting of a positive infant rapid HIV test, confirmatory testing with HIV RNA PCR should be performed on the infant as soon as possible and the infant should immediately be started on three-drug presumptive therapy.
    - If the infant HIV RNA PCR is negative, the positive rapid HIV test result suggests parental HIV infection. However, the infant remains at risk, and the infant's three-drug presumptive therapy should be continued unless parental infection can be excluded.
    - If the infant HIV RNA PCR is positive, this likely represents true HIV infection. HIV RNA PCR should be repeated to confirm the result. Three-drug presumptive therapy should be continued until confirmatory test results are available.

- Discharge of the infant should be delayed until parental HIV status is clarified with rapid HIV testing and/or HIV RNA PCR. Results of HIV RNA PCR testing from infants may take up to a week or more to be available. As long as the infant is on an appropriate antiretroviral regimen based on parental infection status, the infant's HIV RNA PCR result does not need to be available before discharge.

- When a birthing parent declines HIV testing for themself or their infant, providers should have a risk /benefit discussion with the birthing parent, document this in the newborn note, and ask the family to sign the refusal of parental labs consent form as outlined in newborn refusal of care policy. If a birthing parent is known to be at high risk for sexually transmitted diseases (STI) (e.g., IV drug use or past STIs), consultation with Pediatric Infectious Diseases is recommended to discuss potential empiric antiretroviral therapy.

Two-Drug Prophylaxis

- For some infants who do not meet all criteria for being at low risk of transmission but who have few high-risk features, it may be possible to use two-drug prophylaxis with zidovudine and nevirapine instead of three-drug presumptive therapy.

- **Three-drug presumptive treatment remains the recommendation for all high-risk infants and two-drug prophylaxis should only be given in extenuating circumstances**. The decision to pursue this approach should be made in discussion with and under the guidance of the Pediatric Infectious Diseases consult service.

- In this regimen, the infant will receive six weeks of zidovudine as well as three doses of nevirapine in the first week of life. (See Table 2 for nevirapine dosing for this scenario, which differs from dosing used for presumptive therapy.)

Subsequent Postnatal Management of the HIV-Exposed Neonate

- Subsequent postnatal management will typically occur as an outpatient in the Pediatric Infectious Diseases Clinic. However, if an infant remains hospitalized for a prolonged length of stay, the Pediatric Infectious Diseases consult service will assist with ongoing management.

- Virologic tests (listed in MiChart as "HIV RNA PCR") are optimal for diagnosis of HIV infection in infants <18 months of age and should be performed between 14–21 days of life, at 1–2 months of age, and at 4–6 months of age.

- In infants at high-risk who receive presumptive therapy, HIV RNA PCR should also be obtained 2-6 weeks after discontinuation of therapy.

For infants who are being breastfed, additional virologic testing is recommended every three months while breastfeeding is ongoing (including between the tests at 1-2 months of age and 4-6 months of age if the gap between tests is greater than three months) and 4-6 weeks, three months, and six months after cessation of breastfeeding, irrespective of when breastfeeding ends.

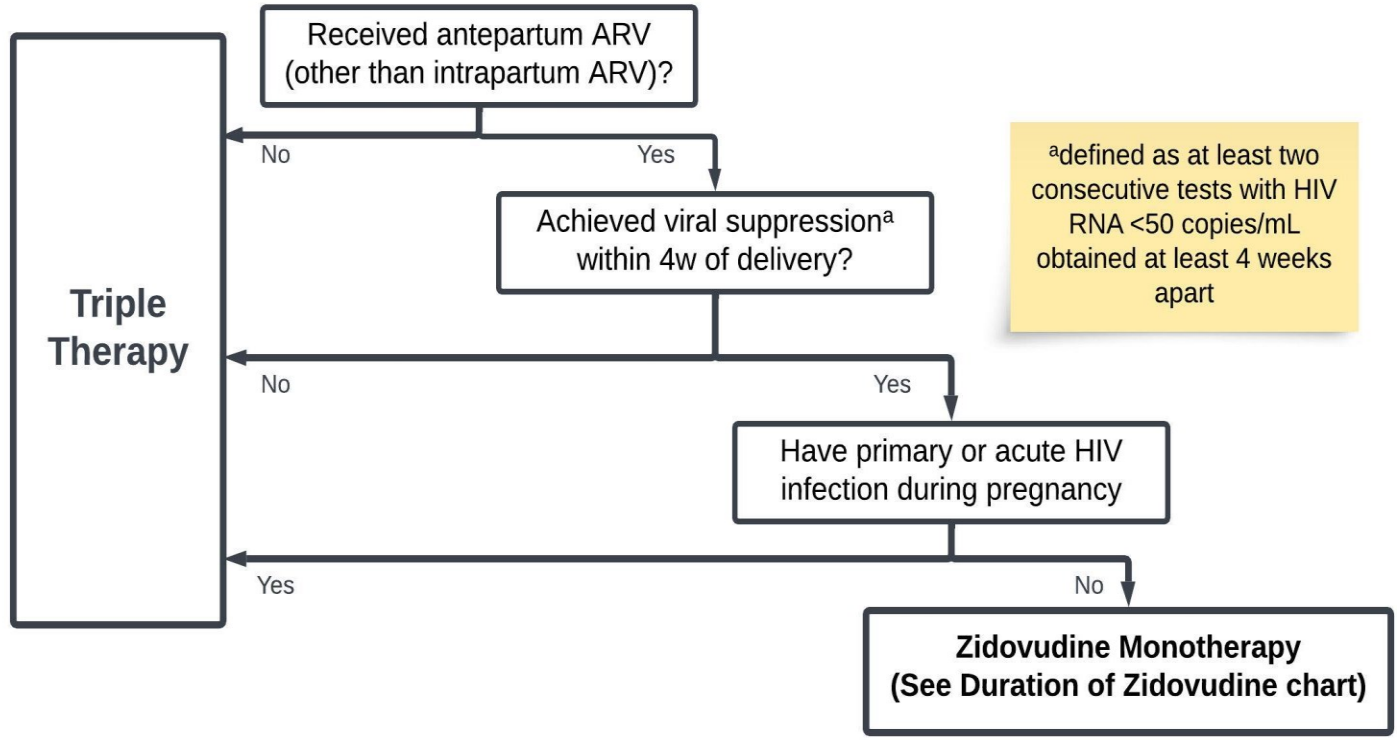
- HIV infection is **presumptively excluded** if two or more virologic tests are negative (one at age ≥2 weeks and one at age ≥4 weeks), if a virologic test at ≥8 weeks is negative, or if an HIV antibody test is negative at ≥6 months.

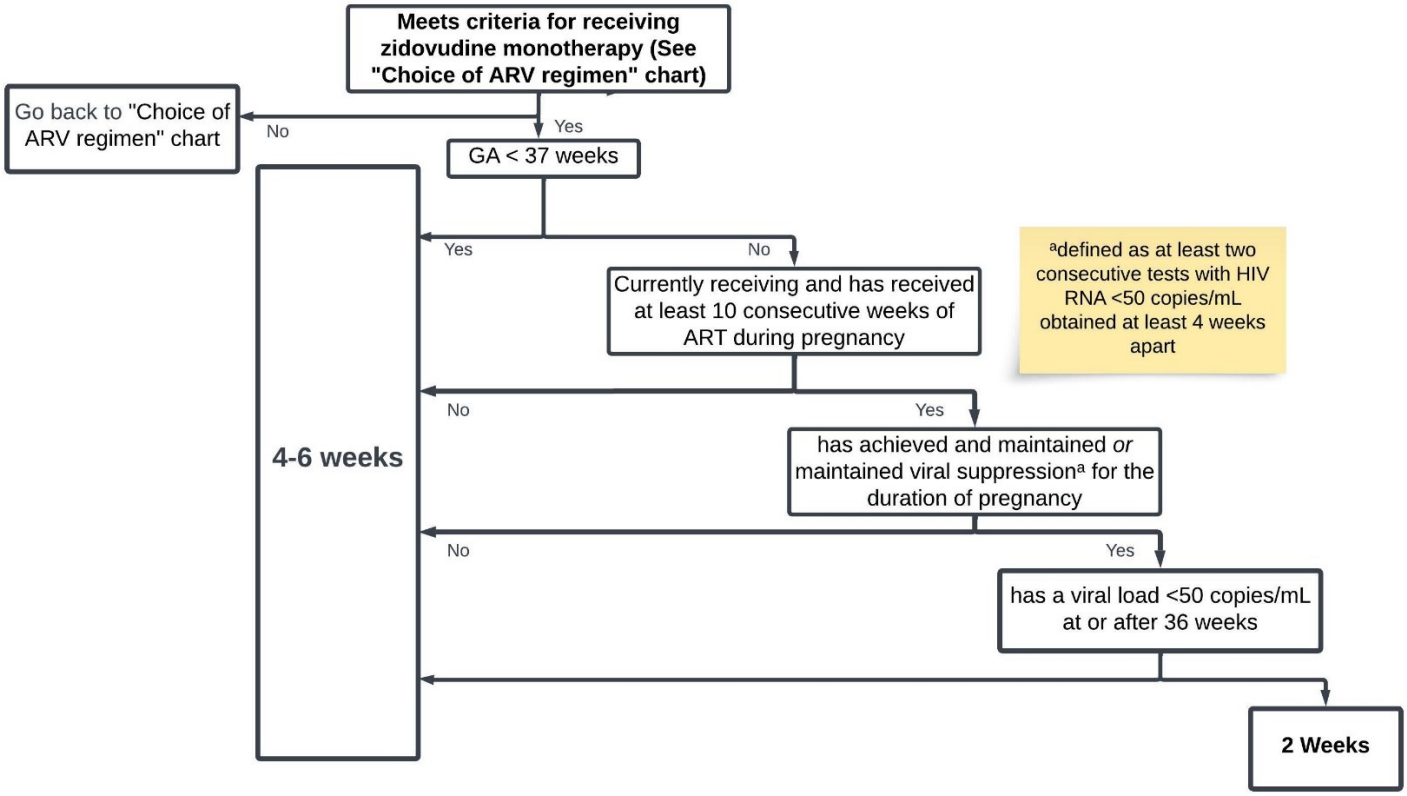
- HIV infection is **definitively excluded** if two or more virologic tests are negative (one at ≥1 month of age and one at ≥4 months of age) or if two HIV antibody tests obtained from separate specimens are negative at ≥6 months. Note that HIV infection does not need to be definitively excluded prior to discontinuing antiretrovirals.

- At a minimum, a CBC with differential should be repeated at 4 weeks of age for any infant who remains on zidovudine or lamivudine. Further decisions about the timing of subsequent monitoring of hematologic parameters depend on baseline hematologic values, gestational age at birth, clinical condition of the infant, the zidovudine dose, concomitant medications, and parental antepartum therapy.

- If hematologic abnormalities are identified in infants receiving antiretroviral prophylaxis, decisions on whether to continue infant antiretroviral prophylaxis should be made on an individual basis in consultation with Pediatric Infectious Diseases.

- To prevent *Pneumocystis jirovecii* pneumonia (PJP), all infants born to individuals with HIV infection should begin PJP prophylaxis at 4 to 6 weeks of age, after completing their antiretroviral prophylaxis regimen, unless HIV infection has been presumptively excluded with two or more negative virologic tests (one at age ≥2 weeks and one at age ≥4 weeks).


Triple Therapy
Received antepartum ARV (other than intrapartum ARV)?
No
Yes
Achieved viral suppression[a] within 4w of delivery?
No
Yes
Have primary or acute HIV infection during pregnancy
Yes
No
Zidovudine Monotherapy (See Duration of Zidovudine chart)
[a]defined as at least two consecutive tests with HIV RNA <50 copies/mL obtained at least 4 weeks apart


Meets criteria for receiving zidovudine monotherapy (See "Choice of ARV regimen" chart)
Go back to "Choice of ARV regimen" chart
No
Yes
GA < 37 weeks
Yes
No
Currently receiving and has received at least 10 consecutive weeks of ART during pregnancy
No
Yes
has achieved and maintained *or* maintained viral suppression[a] for the duration of pregnancy
No
Yes
has a viral load <50 copies/mL at or after 36 weeks
4-6 weeks
2 Weeks
[a]defined as at least two consecutive tests with HIV RNA <50 copies/mL obtained at least 4 weeks apart

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VI: TABLES

Table 1: Antiretroviral Regimens for the Prevention of Perinatal Transmission of HIV

Gestational Age	<32 weeks	≥32 to <34 weeks		≥34 to <37 weeks		≥37 weeks	
Birthweight	**Any weight**	**< 1.5 kg**	**≥ 1.5 kg**	**< 1.5 kg**	**≥ 1.5 kg**	**< 2 kg**	**≥ 2 kg**
Low Risk	Zidovudine Monotherapy						
High Risk Presumptive Regimen[a]	Zidovudine Monotherapy	Zidovudine + Lamivudine	Zidovudine + Lamivudine + Nevirapine	Zidovudine + Lamivudine	Zidovudine + Lamivudine + Nevirapine	Zidovudine + Lamivudine + Nevirapine	Zidovudine + Lamivudine + Raltegravir
High Risk 2-Drug Alternative Regimen	Zidovudine + Nevirapine (prophylactic dosing)						

[a] Consider discussing choice of regimen with the National Perinatal HIV hotline, particularly for infants who are unable to receive raltegravir.

Table 2: Antiretroviral Dosing for the Prevention of Perinatal Transmission of HIV[a,b]

Drug	Gestational Age	Postnatal Age	Weight	Dose	Frequency
Lamivudine	≥32 weeks	Birth-4 weeks		2 mg/kg/dose	Q12h
		4-6 weeks		4 mg/kg/dose	Q12h
Nevirapine[c] (prophylactic therapy)	ALL	ALL	1.5-2 kg	8 mg dose	1st dose: within 48h birth 2nd dose: 48h after 1st dose 3rd dose: 96h after 2nd dose
			>2 kg	12 mg dose	
Nevirapine[c] (presumptive therapy)	≥32 to <34 weeks	Birth-2 weeks		2 mg/kg/dose	Q12h
		2-4 weeks		4 mg/kg/dose	Q12h
		4-6 weeks		6 mg/kg/dose	Q12h
	≥34 to <37 weeks	Birth-1 week		4 mg/kg/dose	Q12h
		1-6 weeks		6 mg/kg/dose	Q12h
	≥37 weeks	Birth-6 weeks		6 mg/kg/dose	Q12h
Raltegravir[d,e]	≥37 weeks	Birth-1 week		1.5 mg/kg/dose	Q24h
		1-4 weeks		3 mg/kg/dose	Q12h
		4-6 weeks		6 mg/kg/dose	Q12h
Zidovudine	<30 weeks	Birth-4 weeks		2 mg/kg/dose	Q12h
		4-6 weeks		3 mg/kg/dose	Q12h
	≥30 to <35 weeks	Birth-2 weeks		2 mg/kg/dose	Q12h
		2-6 weeks		3 mg/kg/dose	Q12h
	≥35 weeks	Birth-6 weeks		4 mg/kg/dose	Q12h

[a] A two-week course of zidovudine is recommended for term, low-risk infants; a 4-6 week course of zidovudine is recommended for low-risk infants <37 weeks gestational age; and a six week course of zidovudine is recommended for high-risk infants. For high-risk infants on three-drug presumptive therapy, the duration of lamivudine and nevirapine or raltegravir may be up to six weeks and will be determined during outpatient follow-up in the Pediatric Infectious Diseases Clinic

[b] Doses provided are for oral/enteral formulations. If intravenous formulation is required, please discuss dosing adjustments with pharmacist

[c] Not recommended for infants <1.5kg.

[d] Not recommended for infants <2kg.

[e] If the mother has taken raltegravir or raltegravir-containing medication within 2 to 24 hours prior to delivery, the neonate's first dose of **raltegravir only** should be delayed until 24 to 48 hours after birth

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References:

- Panel on Treatment of HIV During Pregnancy and Prevention of Perinatal Transmission. Recommendations for the Use of Antiretroviral Drugs During Pregnancy and Interventions to Reduce Perinatal HIV Transmission in the United States. January 31, 2023; pp 1-637.
    - Available at https://clinicalinfo.hiv.gov/sites/default/files/guidelines/documents/perinatal-hiv/guidelines-perinatal.pdf

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10/2024: Removed gendered language referring to birthing parents, updated recommendations for shared decision making regarding breastfeeding, and adjusted treatment durations for low-risk infants to align with most recent national guidelines
C&W Perinatal Joint Practice Committee Approval: 03/2018

The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medical advice based on clinical judgment, or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time to time change. The authors of these guidelines have made all attempts to ensure the accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source.

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